

Sensor-Based Simulation

IoT Temperature Monitoring System

CodeAlpha IoT Internship — Task 2
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Simulation Platform: Tinkercad Circuits

1. Project Overview

This task presents a simulated IoT system that reads ambient temperature using an LM35 analog temperature sensor connected to an Arduino Uno microcontroller. Based on the temperature reading, the system controls two LEDs to indicate whether the temperature is within a safe range or has exceeded a critical threshold — simulating a basic IoT environment monitoring node. The simulation was built and tested using Tinkercad Circuits, a free browser-based electronics simulation platform by Autodesk.

2. Components Used

Component	Quantity	Role
Arduino Uno R3	1	Microcontroller — reads sensor & controls LEDs
LM35 Temperature Sensor	1	Analog temperature measurement (10 mV/°C)
Green LED	1	Indicates normal temperature (< 30°C)
Red LED	1	Indicates high temperature alert (>= 30°C)
220Ω Resistor	2	Current limiting for LEDs
Breadboard	1	Circuit assembly platform
Jumper Wires	Several	Electrical connections

3. Circuit Connections

LM35 Temperature Sensor:

The LM35 has three pins: VCC (+5V), Output (analog voltage), and GND. The VCC pin connects to the Arduino 5V supply. The Output pin connects to Arduino analog pin A0. The GND pin connects to Arduino GND. The LM35 produces a linear output of 10 mV per degree Celsius — for example, 250 mV at 25°C and 300 mV at 30°C.

Green LED (Normal Indicator):

The anode (longer leg) of the green LED connects through a 220Ω resistor to Arduino digital pin 8. The cathode (shorter leg) connects to GND. This LED lights up when the temperature is below 30°C.

Red LED (Alert Indicator):

The anode of the red LED connects through a 220Ω resistor to Arduino digital pin 9. The cathode connects to GND. This LED lights up when the temperature reaches or exceeds 30°C, simulating a temperature alert in an IoT monitoring system.

Wiring Summary Table:

From	To
LM35 — VCC	Arduino 5V
LM35 — Output	Arduino A0
LM35 — GND	Arduino GND
Green LED Anode → 220Ω	Arduino Pin 8
Green LED Cathode	Arduino GND
Red LED Anode → 220Ω	Arduino Pin 9
Red LED Cathode	Arduino GND

4. Arduino Source Code

```
// =====  
//  IoT Temperature Monitoring System – Tinkercad Simulation  
//  CodeAlpha IoT Internship | Task 2  
//  Author: Seif Eldaby  
//  =====  
  
// Pin Definitions  
const int TEMP_SENSOR_PIN = A0;    // LM35 output -> Arduino A0  
const int GREEN_LED_PIN   = 8;     // Normal temperature indicator  
const int RED_LED_PIN     = 9;     // High temperature alert  
  
// Threshold  
const float TEMP_THRESHOLD = 30.0; // degrees Celsius  
  
void setup() {  
  pinMode(GREEN_LED_PIN, OUTPUT);  
  pinMode(RED_LED_PIN,   OUTPUT);  
  Serial.begin(9600);  
  Serial.println("=== IoT Temperature Monitor Started ===");  
}  
  
void loop() {
```

```

// Step 1: Read raw ADC value from LM35 (0 - 1023)
int rawValue = analogRead(TEMP_SENSOR_PIN);

// Step 2: Convert ADC value to voltage (mV)
// Arduino ADC: 10-bit -> 1024 steps over 5000 mV
float voltage_mV = (rawValue / 1024.0) * 5000.0;

// Step 3: Convert voltage to temperature
// LM35 output: 10 mV per degree Celsius
float temperature = voltage_mV / 10.0;

// Step 4: Print to Serial Monitor
Serial.print("Raw ADC: ");
Serial.print(rawValue);
Serial.print(" | Voltage: ");
Serial.print(voltage_mV, 1);
Serial.print(" mV | Temperature: ");
Serial.print(temperature, 1);
Serial.println(" C");

// Step 5: Control LEDs based on temperature
if (temperature >= TEMP_THRESHOLD) {
    digitalWrite(GREEN_LED_PIN, LOW); // Turn OFF green LED
    digitalWrite(RED_LED_PIN, HIGH); // Turn ON red LED (ALERT)
    Serial.println(">> ALERT: High Temperature Detected!");
} else {
    digitalWrite(GREEN_LED_PIN, HIGH); // Turn ON green LED (Normal)
    digitalWrite(RED_LED_PIN, LOW); // Turn OFF red LED
    Serial.println(">> Status: Temperature Normal.");
}

delay(1000); // Wait 1 second before next reading
}

```

5. Code Explanation

5.1 Pin Definitions and Constants

The code begins by defining the pin assignments and the temperature threshold as named constants. Using named constants (rather than hard-coded numbers) makes the code readable and easy to modify. TEMP_THRESHOLD is set to 30.0°C — the point above which the red alert LED activates.

5.2 setup() Function

The `setup()` function runs once when the Arduino powers on. It configures the LED pins as outputs using `pinMode()`, and initializes serial communication at 9600 baud so that temperature readings can be monitored in real time through the Tinkercad Serial Monitor.

5.3 Analog-to-Temperature Conversion

The LM35 outputs an analog voltage proportional to temperature. The Arduino's 10-bit ADC maps this voltage to an integer between 0 and 1023. The code converts this raw value to millivolts by multiplying by (5000 / 1024). Since the LM35 produces exactly 10 mV per degree Celsius, dividing the millivolt value by 10 gives the temperature in Celsius. For example: a raw ADC reading of 61 corresponds to approximately 298 mV, which equals 29.8°C.

5.4 LED Control Logic

An if-else condition compares the calculated temperature against the threshold. If the temperature is at or above 30°C, the green LED turns off and the red LED turns on, simulating an IoT alert notification. If the temperature is below the threshold, the green LED remains on and the red LED stays off, indicating normal operating conditions. The `delay(1000)` at the end of the loop introduces a 1-second pause between readings, appropriate for a real-time monitoring application.

6. How to Run This Simulation in Tinkercad

Step	Action
1	Go to tinkercad.com and sign in (free account)
2	Click "Create" → "Circuit"
3	Search and add: Arduino Uno, LM35, 2x LED (green & red), 2x 220Ω Resistor, Breadboard
4	Connect components exactly as described in the Wiring Summary Table (Section 3)
5	Click "Code" → switch to "Text" mode → paste the Arduino code from Section 4
6	Click "Start Simulation"
7	Open Serial Monitor to view live temperature readings
8	Click the LM35 sensor and adjust the temperature slider to test LED switching
9	Take screenshots of both states: Normal (green ON) and Alert (red ON)

7. Expected Simulation Output

Temperature Reading	Green LED	Red LED	Serial Monitor Message
< 30°C (e.g. 25°C)	ON	OFF	Status: Temperature Normal.
>= 30°C (e.g. 35°C)	OFF	ON	ALERT: High Temperature Detected!

The Serial Monitor will display continuous output in the following format:

```
=== IoT Temperature Monitor Started ===  
Raw ADC: 51 | Voltage: 249.0 mV | Temperature: 24.9 C  
>> Status: Temperature Normal.  
Raw ADC: 51 | Voltage: 249.0 mV | Temperature: 24.9 C  
>> Status: Temperature Normal.  
Raw ADC: 61 | Voltage: 298.1 mV | Temperature: 29.8 C  
>> Status: Temperature Normal.  
Raw ADC: 62 | Voltage: 302.7 mV | Temperature: 30.3 C  
>> ALERT: High Temperature Detected!
```

8. IoT Context and Real-World Application

In a real-world IoT deployment, this simulation would be extended with several additional capabilities. The Arduino would be replaced or augmented with a Wi-Fi or GSM module (such as an ESP8266 or SIM800L) to transmit temperature data to a cloud platform like ThingSpeak, Firebase, or AWS IoT Core. Instead of local LEDs, alerts would be sent as push notifications to a mobile app or as emails/SMS messages to facility managers. Multiple sensor nodes would be deployed across a building or industrial site, all reporting to a central dashboard for unified monitoring.

This simulation therefore represents the fundamental sensing and logic layer of a complete IoT system — demonstrating the core principles of data acquisition, threshold-based decision making, and actuator control that underpin all real-world IoT applications.

References

- [1] Texas Instruments. (2017). LM35 Precision Centigrade Temperature Sensor Datasheet. TI Documentation.
- [2] Arduino. (2023). `analogRead()` Reference Documentation.
arduino.cc/reference/en/language/functions/analog-io/analogread/
- [3] Autodesk Tinkercad. (2024). Tinkercad Circuits — Browser-Based Electronics Simulation.
tinkercad.com
- [4] Monk, S. (2016). Programming Arduino: Getting Started with Sketches (2nd ed.). McGraw-Hill Education.